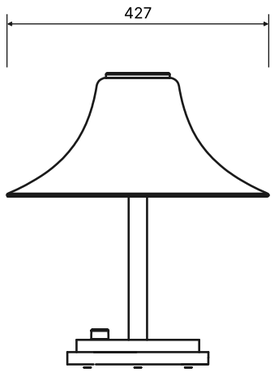
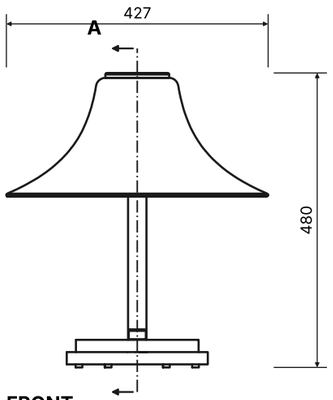
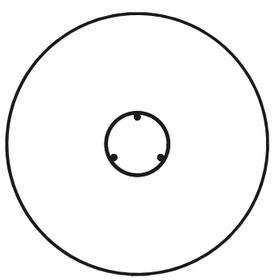
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT A



FROM ABOVE

General arrangement

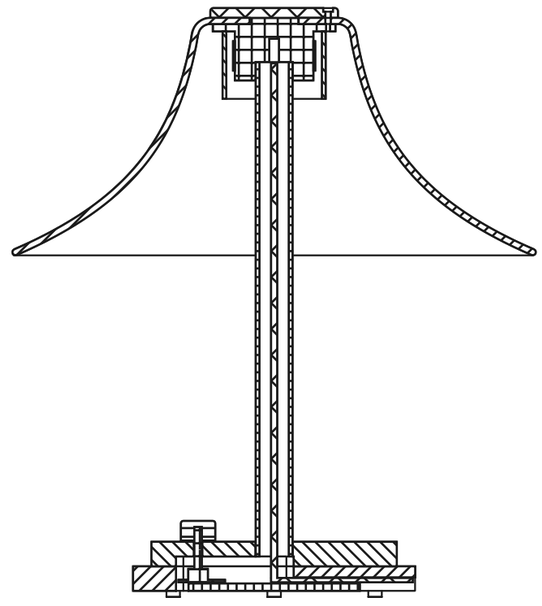
Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

EL-001 rev A

Scale 1:5. Section A-A (sheet 1): the product cut at x 216.0 and seen from the right; cut faces hatched, a different angle for each part.



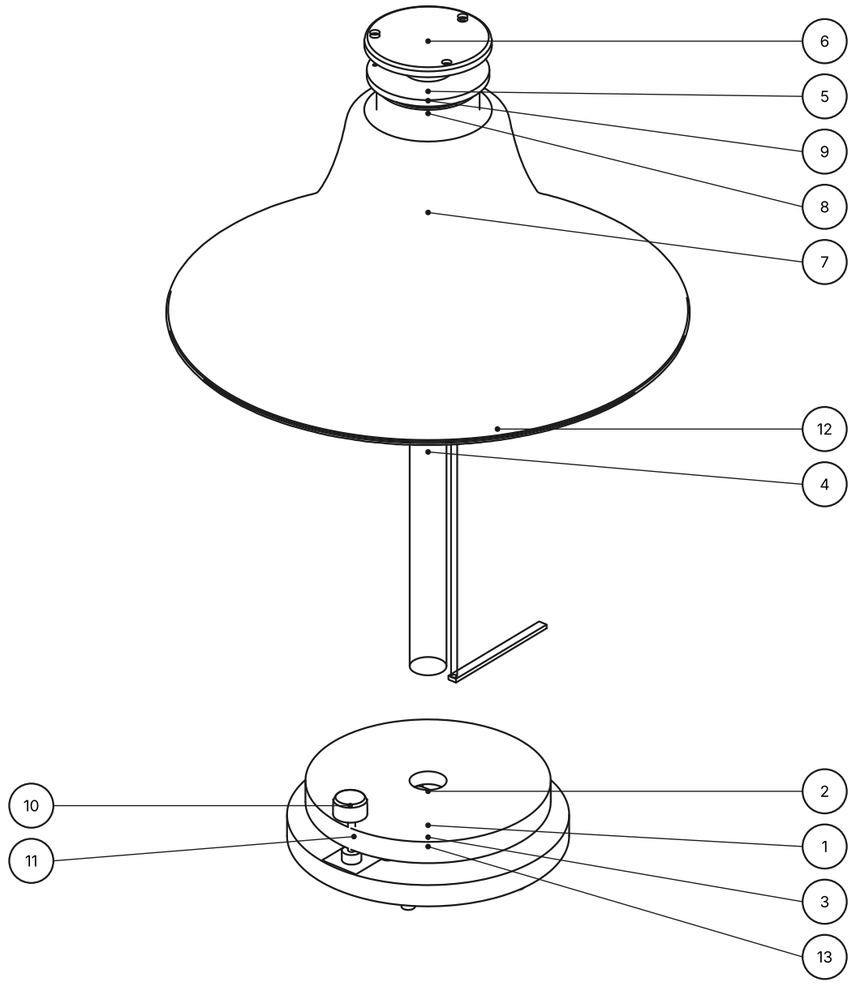
Section A-A

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5, parts drawn apart. Balloons are the parts list's numbers (last sheet).



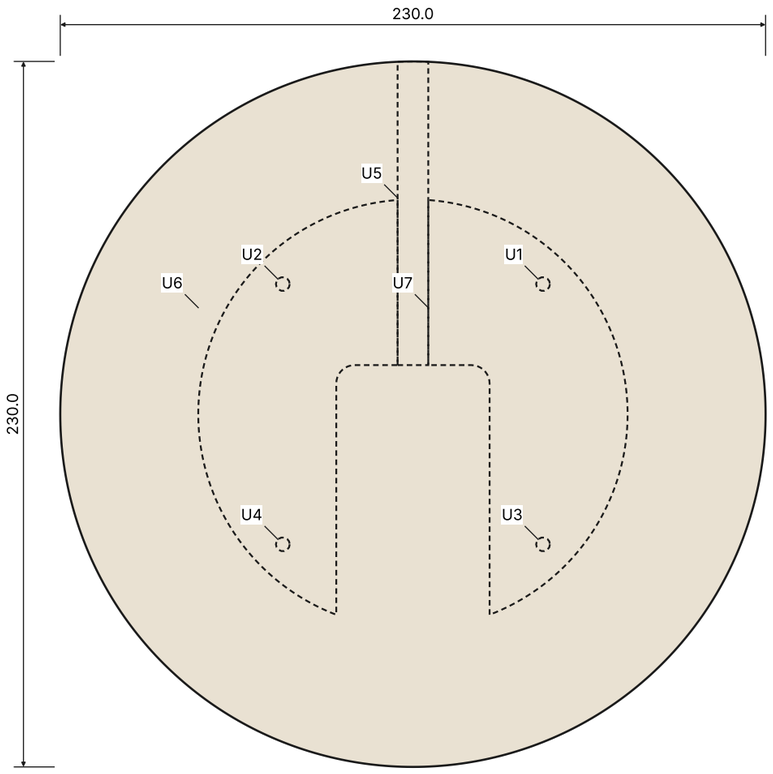
Exploded view

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/foot-lower.dxf. Marks listed at the right.



mark	size	x	y	depth
U1	Ø4.5	157.4	157.4	14 from inside
U2	Ø4.5	72.6	157.4	14 from inside
U3	Ø4.5	157.4	72.6	14 from inside
U4	Ø4.5	72.6	72.6	14 from inside
U5	profile (DXF)	115.0	180.5	11 from inside
U6	profile (DXF)	77.5	117.2	6 from inside
U7	profile (DXF)	152.5	117.2	6 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

foot-lower: 20 mm cast-acrylic, 1 for the set, seen from its outer face
inner-face pockets: dxf/foot-lower-underside.dxf (seen from that face)

Sheet part: foot-lower

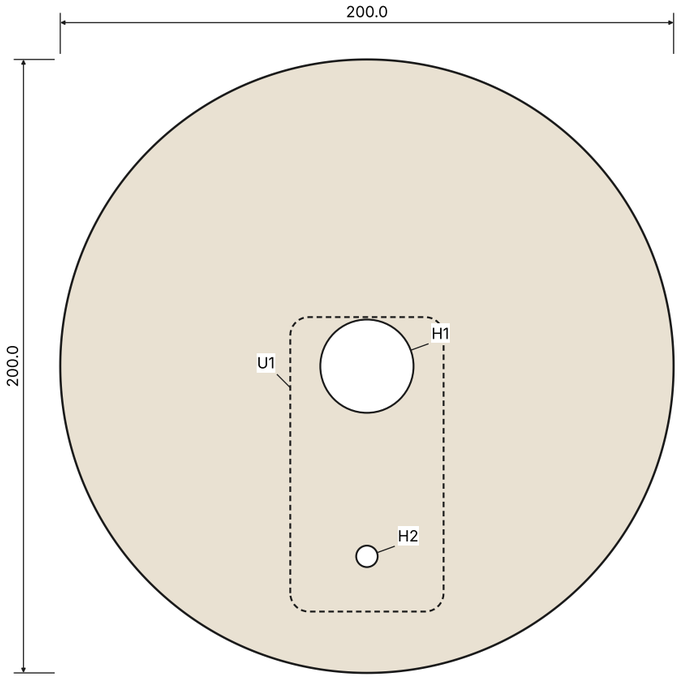
Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:2. As the router sees it from the outer face; DXF: dxf/foot-upper.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	Ø30.4	100.0	100.0	through
H2	Ø7.0	100.0	38.0	through
U1	profile (DXF)	100.0	68.0	8 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



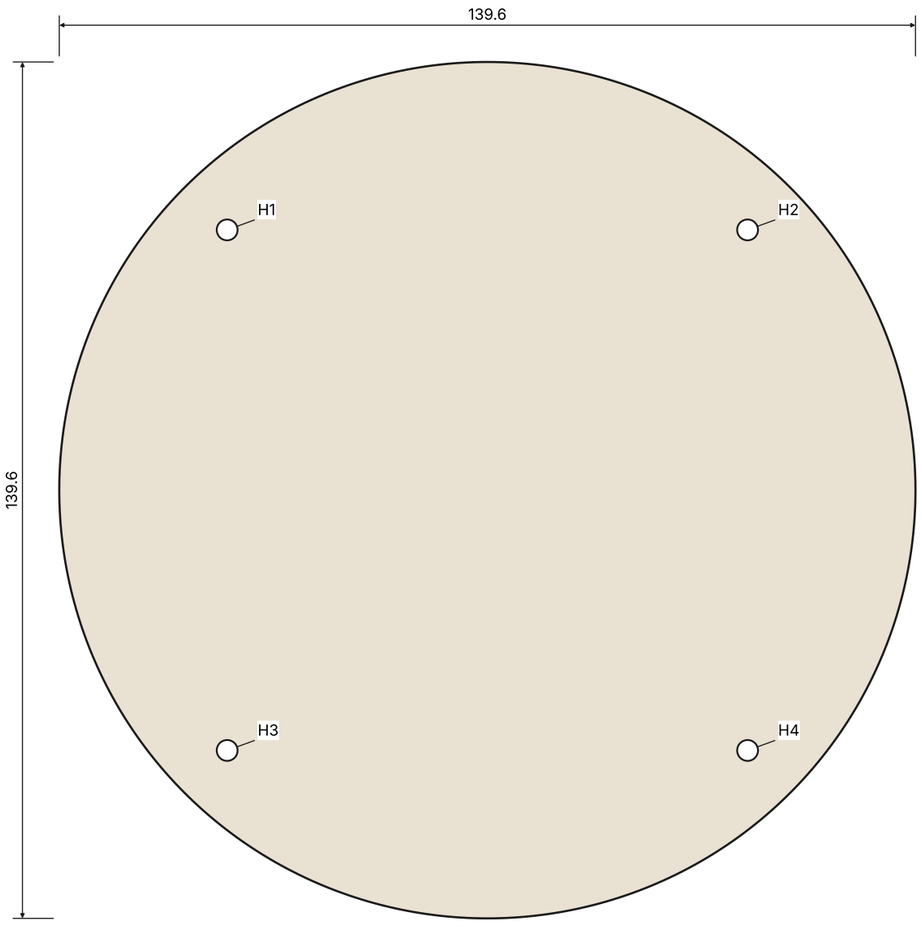
foot-upper: 20 mm cast-acrylic, 1 for the set, seen from its outer face
inner-face pockets: dxf/foot-upper-underside.dxf (seen from that face)

Sheet part: foot-upper

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:1. As the router sees it from the outer face; DXF: dxf/cover.dxf. Marks listed at the right.



cover: 6 mm cast-acrylic, 1 for the set, seen from its outer face

mark	size	x	y	depth
H1	Ø3.4	27.4	112.2	through
H2	Ø3.4	112.2	112.2	through
H3	Ø3.4	27.4	27.4	through
H4	Ø3.4	112.2	27.4	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: cover

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

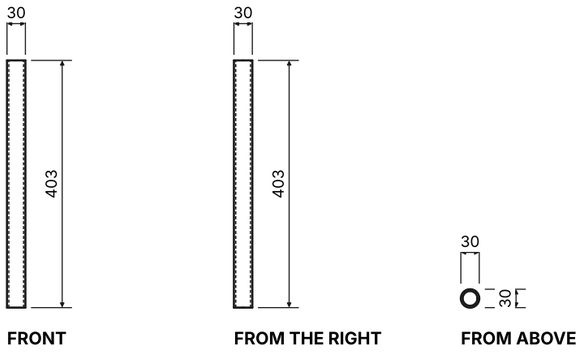
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

4. stem: lathe, cast-acrylic; 103 cm3 each; 1 for the set; 1:10

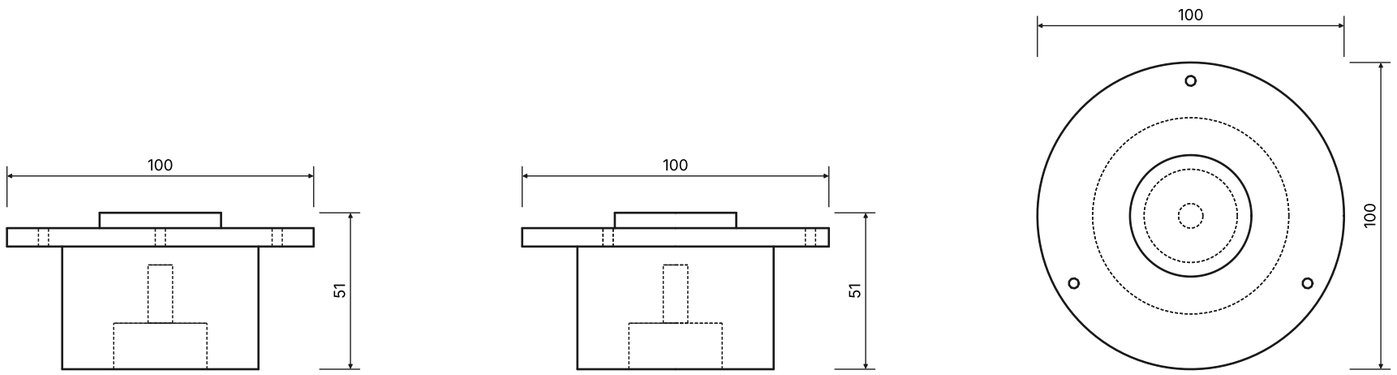
cast acrylic tube 30 × 3, cut square and its ends polished; bonded into the foot's socket and the hub's bore



FRONT **FROM THE RIGHT** **FROM ABOVE**

5. hub: lathe, aluminium; 170 cm3 each; 1 for the set; 1:2

6061, bead blasted, clear anodised: the strip's heat sink; three M4 tapped in its flange



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

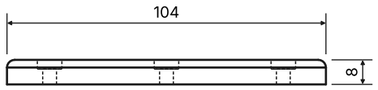
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

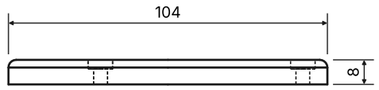
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

6. cap: lathe, aluminium; 67 cm3 each; 1 for the set; 1:2

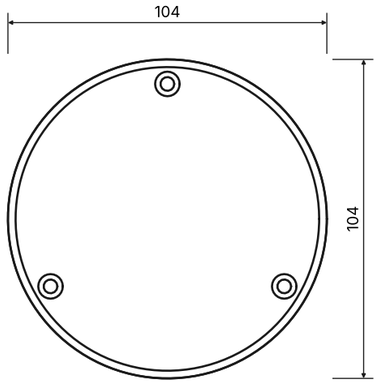
clamps the shade's collar on the hub's flange, three M4 button heads counterbored



FRONT



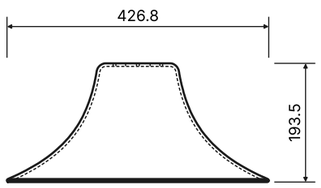
FROM THE RIGHT



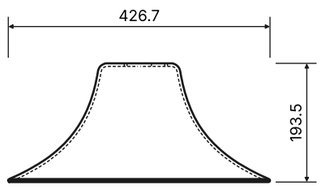
FROM ABOVE

7. shade: formed, cast-acrylic-formed; 1001 cm3 each; 1 for the set; 1:10

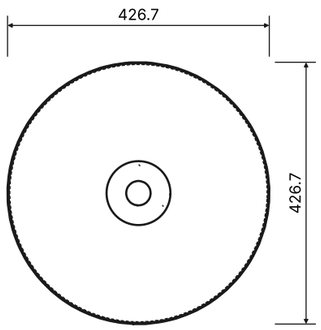
the buck about $900, a tenth on each of a first run of ten (the setup here); formed from 5 mm cast sheet tinted four times as dense as the cabinets' 20 mm, so it reads the same, over a turned buck (about 160 C), annealed; its rim trimmed and rolled R2.5, its collar machined flat and drilled for three M4



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

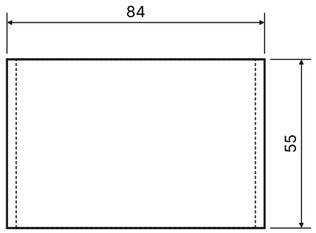
mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

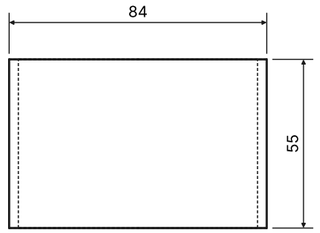
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. sleeve: lathe, cast-acrylic; 42 cm3 each; 1 for the set; 1:2

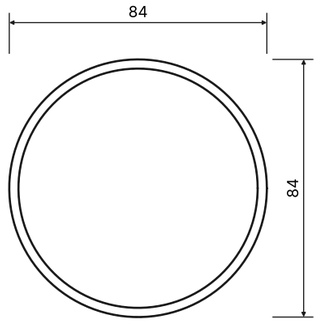
opal (white diffusing) cast acrylic tube 84 × 3, bonded to the flange's underside



FRONT



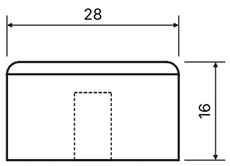
FROM THE RIGHT



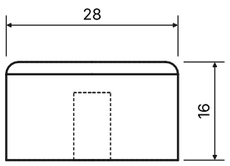
FROM ABOVE

10. knob: lathe, cast-acrylic; 9 cm3 each; 1 for the set; 1:1

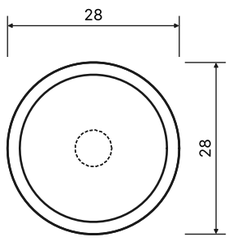
turned from rod in the colour, deeper; its edge eased; a 6 bore for the pot's shaft, set screw from below



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 1; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	foot-lower	cut	1	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	230 × 230	25.13
2	foot-upper	cut	1	20 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	200 × 200	19.00
3	cover	cut	1	6 mm Cast acrylic, tinted through in the speaker's colour, 4 × 8 ft sheet	140 × 140	2.78
4	stem	lathe	1	Cast acrylic tube, tinted through	40 × 40 × 413	71.43
5	hub	lathe	1	Aluminium 6061	110 × 110 × 61	63.45
6	cap	lathe	1	Aluminium 6061	114 × 114 × 18	50.85
7	shade	formed	1	Cast acrylic sheet 3 mm, tinted through, drape-formed over a buck	1001 cm3, 437 × 437 × 203	139.04
8	sleeve	lathe	1	Cast acrylic tube, tinted through	94 × 94 × 65	67.97
9	strip	buy	1	Waveform Lighting CENTRIC WARM LED strip, 24 V, 95+ CRI, 2700 K at full and dimming warmer toward 2000 K (PN 3008.27), 450 lm a foot; 0.40 m a lamp (two turns round the hub, 64 across, cut at its marks)		[PRICE]
10	knob	lathe	1	Cast acrylic tube, tinted through	38 × 38 × 26	46.50
11	dimmer	buy	1	LEDSupply Single Color Mini Rotary Dimmer, 12-24 V DC PWM, with an on/off switch at the end of its turn, 2.1 mm plug in; its board taken out of the housing and screwed into the foot's pocket, the knob replaced by the acrylic one		9.00
12	cord	buy	1	Satco / Nuvo 18/2 SPT-2 lamp cord, clear (80-2374), 8 ft: from the foot's back to a 2.1 mm in-line DC socket for the adapter, and on up the stem to the hub (low voltage only)		7.00
13	bumpons	buy	1	3M Bumpon SJ5302 clear hemispherical bumpers 11 × 5, a set of 6 (under the foot)		5.00
14	adapter	buy	1	Mean Well GST25U24-P1J wall-plug adapter, 24 V DC 1.04 A (25 W), level VI, its 2.1 mm plug on a 1.5 m lead; limited power source by one listing		18.00
15	dc-socket	buy	1	In-line 2.1 × 5.5 mm DC socket with strain relief, for the cord's end		3.00
16	cord-grip	buy	1	Clear nylon strain-relief bush for 18/2 SPT-2 cord in a 9.5 hole (Heyco kind)		2.00
17	fixings	buy	1	The lamp's screws: three M4 × 12 black button heads and M4 brass inserts bonded in the hub's flange (the shade's collar), four M3 × 8 and inserts (the foot's cover), two M3 × 6 (the dimmer's board), black nylon washers		6.00
18	thermal	buy	1	The strip's own adhesive on the hub, and 3M 8810 thermally conductive tape at the hub's bore round the cord's exit		4.00
19	collar-gasket	buy	1	Silicone ring gasket 1 mm, cut to the shade's collar (68 × 92), so the formed collar is clamped without stress		3.00
20	cement	buy	1	Two-part polymerising acrylic cement for cast sheet (Acrifix 1R 0192 kind), clear: the cemented layers (a 100 g kit does one piece)		20.00

Parts list

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Notes

1. The shade: 5 mm tinted cast acrylic (the Clouds' own) formed over a turned buck, about 160 C, annealed after; its rim trimmed and rolled, its collar machined flat. It hangs from its collar, clamped between the hub's flange and the cap on a silicone ring, three M4 screws: nothing presses on the formed wall.
2. The light: 0.4 m of 24 V LED strip, 95 CRI, dimming from 2700 K toward 2000 K, wound twice round the hub on its own adhesive; the hub (aluminium, clear anodised) is its heat sink; an opal acrylic sleeve round it so no dot shows. About 600 lm at full, 6 W or so (confirm on the strip's datasheet).
3. Power: a listed 24 V DC wall adapter (Mean Well GST25U24), a 2.1 mm in-line socket on a clear 18/2 cord into the foot's back; the dimmer (12-24 V PWM, its switch at the end of the turn) in the foot's pocket; the cord on up the stem to the hub. Nothing in the lamp is at mains voltage.
4. The foot: two 20 mm discs, 230 and 200, cemented and annealed, the edges eased R2 and polished; the pocket through the lower closed by a 6 mm cover on four M3 screws into bonded inserts; six clear bumpers under it.
5. The stem: cast acrylic tube 30 × 3 bonded into the foot's socket and the hub's bore with a structural acrylic adhesive.
6. Stability: about 1.7 kg in the foot and 1.6 kg in the head; it tips past about 28 degrees, as a table lamp of its size should.

Notes

Bell: Table lamp in tinted cast acrylic: the horn's flare as its shade

mm | issue 2026-10-18 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15